SECJ 4383 – Software Constructions Lab Exercise

Lab Exercise 1b – DevOps Continous Integration Lab

(This lab exercise is not self-explanatory. Further explanation will be given by the instructor during the lab session)

Devops Lab Instruction (Detailed Instructions in the video)

Part 1: Continuous Integration (CI)

1. Login to Jenkins > Manage Plugins > Available Plugins > Search “Github Servers” or anything related to github > Install Plugin
2. Jenkins Dashboard > Manage Jenkins > Configure > Github > Github Servers > Advanced > Check “Override hook url” > Copy URL
3. Login to github account > Current repository > Setting > Webhook > Add webhook > Insert copied URL (Make sure to change localhost to public ip)
4. Jenkins Dashboard > Job > Configure > Build triggers > Check “Github hook trigger for GITScm Polling” to receive any changes from github

Part 2:Continuous Deployment

1. Download Docker Desktop at https://www.docker.com/products/docker-desktop/ and finish installation
2. Go to https://hub.docker.com/ and create account
3. Login docker desktop using docker hub account
4. Download minikube (Kubernetes) at https://minikube.sigs.k8s.io/docs/start/. Follow installation instructions
5. Run ‘minikube start’ to ensure kubernetes service is running
6. Run ‘minikube dashboard’ to get kubernetes url, namespace and cluster name
7. Go to spring boot project > Root path > Create Dockerfile (Ensure that project information is correct) > Create deployment.yaml and service.yaml
8. Go to Jenkins > Manage Plugins > Install Docker and kubernetes related plugins
9. Add docker and kubernetes credential in jenkins
10. Go to Jenkins > Add new job > Select pipeline type > Follow instructions from video
11. Pipeline script template can be obtained from ‘pipeline syntax’ from Pipeline section in specific job
12. Commit changes from local to github and observe the automated process